

**Project update:**

The research questions of this project are: 1- How are restaurant prices (price level and price range) associated with neighborhood socioeconomic status (SES), measured by median household income and educational attainment, across Chicago? 2- Beyond price, do restaurant characteristics (such as restaurant type, regular opening hours, service offerings) as well as ratings and review volume vary systematically across Chicago neighborhoods? The project's focus area is Descriptive Spatial Analysis and Maps. I collect restaurant data from the Google Places API using a grid-based sampling strategy constrained by the official Chicago boundary polygon to ensure its geographic coverage. Each restaurant will be geocoded and assigned to a Chicago community area through spatial joins. I will then aggregate restaurant signals within community areas and visualize patterns using maps and descriptive comparisons across SES levels.

**Data status:**

- Google Places API (New)
  - to collect Chicago restaurant data
  - Collected
  - included in the original proposal
- Yelp Fusion API
  - to validate and benchmark Google Maps measures (especially ratings and price) and assess potential platform differences
  - Not collected
  - included in the original proposal
  - The free API tier is limited to 5,000 calls per month, and paid usage would exceed my funding limit. I plan to collect as much relevant information as possible for each restaurant within the free tier
- Chicago Data Portal ACS 5 Year Data by Community Area (SES)
  - neighborhood SES indicators (median household income and education)
  - Downloaded from official website
  - [https://data.cityofchicago.org/Community-Economic-Development/ACS-5-Year-Data-by-Community-Area/t68z-cikk/about\\_data](https://data.cityofchicago.org/Community-Economic-Development/ACS-5-Year-Data-by-Community-Area/t68z-cikk/about_data)
  - included in the original proposal
- Chicago Data Portal Boundaries - Community Areas - Map
  - official polygons for assigning restaurants to community areas
  - Downloaded from official website
  - <https://data.cityofchicago.org/Facilities-Geographic-Boundaries/Boundaries-Community-Areas-Map/cauq-8yn6>
  - included in the original proposal

**Analysis plan update**

- Spatial scatter map: Plot restaurant locations and color points by price level/price range to visualize clustering across Chicago
- Choropleth maps (by census tract/neighborhood): Map restaurant density, average/median price level, and mean rating (and optionally review volume) to compare neighborhood patterns
- SES vs service heatmap: Show the share of restaurants offering delivery/takeout/dine-in/reservable across SES bins (e.g., income or education quartiles)
- Regression models: how neighborhood SES predicts price and ratings (as dependent variables), with controls such as restaurant type/categories and review count

**Timeline:**

- Week 5: data collection
- Week 6: complete data cleaning and conduct initial descriptive analysis
- Week 7 & 8 : complete regression analyses and finalize mapping/visualizations

**Team responsibilities:**

This is an individual project.